

Year 7 • Chapter 5A • Algebra: answers (teacher copy)

5.01 The commutative and associative laws (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: 28
2. Ex 1b: 118
3. Ex 2a: 170
4. Ex 2b: 3 100
5. Ex 3a: 390
6. Ex 3b: 2 800

Easy • Set A (quick drill)

1. 28
2. 118
3. 75
4. 58
5. 117
6. 50
7. 37
8. 119
9. 430
10. 900
11. 81
12. 700
13. 88
14. 130

Easy • Set B

1. 170
2. 3 100
3. 900
4. 1 300
5. 700
6. 400

Medium • Set C

1. 390
2. 2 800
3. 600
4. 2 400
5. 2 400
6. 300

Challenging • Set D

1. No: $10 - 4 = 6$ but $4 - 10 = -6$
2. No: $(24 \div 6) \div 2 = 2$ but $24 \div (6 \div 2) = 8$
3. 17 000
4. 210 (10 pairs of 21)

Extension

1. 90 000
2. $38 + 62 = 100$ and $47 + 53 = 100$

Exit ticket

1. Level 1: 47
2. Level 2: 230
3. Level 3: 1 300

5.02 The distributive law (booklet p. 7)

WE DO examples (teacher)

1. Ex 1a: 198
2. Ex 1b: 136
3. Ex 2a: 297
4. Ex 2b: 156
5. Ex 3a: 288
6. Ex 3b: 3 465

Easy • Set A (quick drill)

1. 84
2. 92
3. 217
4. 95
5. 126
6. 116
7. 80
8. 126
9. 76
10. 96
11. 693
12. 205
13. 162
14. 408

Easy • Set B

1. 198
2. 136
3. 297
4. 156
5. 105
6. 117

Medium • Set C

1. 288
2. 3 465
3. 2 323
4. 636
5. 1 592
6. 1 365

Challenging • Set D

1. $7 \times 100 = 700$
2. $15 \times 10 = 150$
3. e.g. $18 \times 25 = 10 \times 25 + 8 \times 25 = 450$; or $20 \times 25 - 2 \times 25 = 450$
4. $97 = 100 - 3$, so multiply each part by 6

Extension

1. 64 064
2. 3 700

Exit ticket

1. Level 1: 84
2. Level 2: 114
3. Level 3: 4 590

5.03 Variables and algebraic notation

(booklet p. 11)

WE DO examples (teacher)

- Ex 1a: $a - (-a) = 2 \times a$
- Ex 1b: $4x$
- Ex 2a: $10a$
- Ex 2b: $24abc$
- Ex 3a: $4 \times m \times n \times k$
- Ex 3b: $9 \times c \times c$; $6 \times a \div b$

Easy · Set A (quick drill)

- $4x$
- $\frac{u}{5}$
- y
- $10a$
- $7m$
- pq
- $3a$
- $4b$
- $6y$
- mn
- k^2
- $p/3$
- $10xy$
- t

Easy · Set B

- $24abc$
- $20mw^2$
- k^3
- $3y^2$
- $2a + 3b$
- $2d$

Medium · Set C

- $4 \times m \times n \times k$
- $6 \times a \div b$
- $9 \times c \times c$
- $3 \times x \times y \times y$
- $a \times a \times a \times b$
- $5 \times m \div (2 \times n)$

Challenging · Set D

- $a + a = 2 \times a$
- $a \times 0 = 0$
- $3n + 1$ sticks
- No: $3a^2 = 3 \times a \times a$ but $(3a)^2 = 9 \times a \times a$

Extension

- $n \times n - (n - 1)(n + 1) = 1$
- check

Exit ticket

- Level 1: $3y$
- Level 2: $12pq$
- Level 3: $5 \times a \times a \times b$

5.04 Expressions from words (booklet p. 15)

WE DO examples (teacher)

- Ex 1a: $3N$
- Ex 1b: $N - 5$
- Ex 2a: $N + 1$
- Ex 2b: $\frac{N}{3}$
- Ex 3a: $2H - k$
- Ex 3b: $2(x + 4)$

Easy · Set A (quick drill)

- $N + 7$
- $3N$
- $N - 5$
- $\frac{N}{4}$
- $N + 1$
- $\frac{N}{3}$
- $x + 5$
- $x - 8$
- $2x$
- $x/3$
- x^2
- xy

Easy · Set B

- $M + 3$
- $B + 5$
- $2H - k$
- ab
- $y - 10$
- $\frac{p}{6}$

Medium · Set C

- $2(x + 4)$
- t^2
- $3w - 7$
- $\frac{m-1}{2}$
- $n - 1$
- $\frac{a+b}{2}$

Challenging · Set D

- $3n$ dollars
- $2s + 4$
- $2L + 10$
- $50 - 2.5k$

Extension

- $3e + 6$
- $3n + 3 = 3(n + 1)$

Exit ticket

- Level 1: $N + 12$
- Level 2: $y - 4$
- Level 3: $5m + 2$

5.05 Substitution (booklet p. 19)

WE DO examples (teacher)

- Ex 1a: 7
- Ex 1b: 0
- Ex 2a: 60
- Ex 2b: 109
- Ex 3a: -11
- Ex 3b: 94

Easy · Set A (quick drill)

1. 7
2. 4
3. 20
4. 30
5. 13
6. 0
7. 7
8. 6
9. 12
10. 1
11. 20
12. 9
13. 13
14. 6

Easy · Set B

1. 6
2. 60
3. 36
4. 3
5. 14
6. 19

Medium · Set C

1. 109
2. -11
3. 169
4. 94
5. 19
6. 319

Challenging · Set D

1. 22
2. 35
3. 77
4. 3

Extension

1. \$59
2. 60 minutes

Exit ticket

1. Level 1: 20
2. Level 2: 13
3. Level 3: 10

5.07 Adding and subtracting terms (booklet p. 23)

WE DO examples (teacher)

1. Ex 1a: $7m$
2. Ex 1b: $8m$
3. Ex 2a: $2y$
4. Ex 2b: $18uv + 4$
5. Ex 3a: $13h^2 + 17h$
6. Ex 3b: $2h^3 - 14h$

Easy · Set A (quick drill)

1. $7m$
2. $8m$
3. $6a$
4. 0
5. $9x$
6. $2y$
7. $7a$
8. $4x$
9. $3m$
10. $5k$
11. $4p + 2q$
12. $6y$
13. $7ab$
14. 0

Easy · Set B

1. yes
2. no
3. yes
4. no
5. no
6. yes

Medium · Set C

1. $18uv + 4$
2. $7a + 4b$
3. $10x + 4$
4. $6pq$
5. $2m + 6$
6. 5

Challenging · Set D

1. $13h^2 + 17h$
2. $2h^3 - 14h$
3. $8ab + 5a - b$
4. $6x + 5$

Extension

1. $6a^2 + 2a - 6$
2. $6m$

Exit ticket

1. Level 1: $9k$
2. Level 2: $3a + 9$
3. Level 3: $2x^2 - 3x$

5.08 Multiplying terms (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: $12m^2$
2. Ex 1b: a^4
3. Ex 2a: $180v^2$
4. Ex 2b: $15p^2q^2$
5. Ex 3a: $21a^2$
6. Ex 3b: $3m^3n^3$

Easy · Set A (quick drill)

1. $12m$
2. $10a$
3. ab
4. a^4
5. $7xy$
6. $12m^2$
7. $10x$
8. $12a$
9. $2xy$
10. m^2
11. $6pq$
12. $4k^2$
13. $-15y$
14. $8ab$

Easy · Set B

1. $12pq$
2. $180v^2$
3. $2k^2$
4. $10a^2b$
5. $12x^3$
6. $8c^3$

Medium · Set C

1. $21a^2$
2. $-12ab$
3. $-3x^2$
4. $8m^2$
5. $-6x^3y$
6. $5t^2$

Challenging · Set D

1. $15p^2q^2$
2. $3m^3n^3$
3. $6a^5$
4. $6xy \text{ cm}^2$

Extension

1. $9m^2$
2. $5xy$

Exit ticket

1. Level 1: $15k$
2. Level 2: $24a^2b$
3. Level 3: $6x^2y$

5.09 Dividing terms (booklet p. 31)**WE DO examples (teacher)**

1. Ex 1a: $5m$
2. Ex 1b: 9
3. Ex 2a: $7y$
4. Ex 2b: $3x$
5. Ex 3a: $\frac{16e}{5}$
6. Ex 3b: $2xy$

Easy · Set A (quick drill)

1. $5m$
2. 9
3. $7y$
4. $4ab$
5. 4
6. $6p$
7. $3a$
8. $3m$
9. 10
10. $3q$
11. $2a$
12. 3

Easy · Set B

1. $3x$
2. $3m$
3. $2a$
4. $3t^2$
5. $\frac{x}{4}$
6. $\frac{2}{3b}$

Medium · Set C

1. $\frac{16e}{5}$
2. $-5n$
3. $3x$
4. $3ab$
5. $\frac{2q}{3}$
6. $-9y$

Challenging · Set D

1. $2xy$
2. 3
3. m
4. $4y \text{ cm}$

Extension

1. $\frac{3m}{2}$
2. $2y$

Exit ticket

1. Level 1: $3p$
2. Level 2: $4ab$
3. Level 3: $\frac{-2x^2}{3}$